पा

पाथीभरा लेखा प्रणाली

Pathibhara Accounting System

Complete Guide: Workflow, Features, Rules and Technical Design
पाथीभरा क्षेत्र विकास समिति, ताप्लेजुङ

Who this is for	Office head (Super Admin), accountant, auditor, and any developer who maintains the code
System version	Phase 1 + Income (आम्दानी), commit `bda80d2`
Fiscal year set up	आ.व. 2082/83 (2082.04.01 to 2083.03.end)
Source code	github.com/nabinbaral01/lekha

Contents

1. What the system does: the one big idea
2. Users and what each one can do
3. Starting, stopping and sharing the site
4. Map of every screen
5. First-time setup (Super Admin)
6. Daily work: entering a payment voucher
7. Voucher life: draft → submit → approve → paid
8. Tax deducted at source (TDS) and कर जम्मा
9. Income (आम्दानी)
10. Books and reports produced automatically
11. Monthly and year-end routine
12. Every safety rule and error message
13. Audit trail and security
14. Technical design (for developers)
15. Database tables
16. API reference
17. Testing
18. Troubleshooting
19. What is still to be built, and what must be verified

1. What the system does: the one big idea

This is a cash-basis government accounting system that follows the म.ले.प. (Auditor General) formats. It replaces the paper books the office keeps today.

> **The golden rule: enter each transaction once.** Every transaction is entered once, as a **गोश्वारा भौचर (Form 203)**. When the office head approves it, the system writes it into **every book automatically**: the bank-cash book, the budget statement, the tax register, the income report, the dashboard charts and the audit log. No book is typed separately, so the books can never disagree with each other.

The four funds (कोष)

Money is kept in four separate funds. Every voucher belongs to exactly one fund.

	पूँजीगत (Capital): assets, construction, purchases	**चालु** (Recurrent): operations, services, programmes
आन्तरिक (Internal): the committee's own revenue	आन्तरिक – पूँजीगत	आन्तरिक – चालु
अनुदान (Grant): money from government or other bodies	अनुदान – पूँजीगत	अनुदान – चालु

The four funds share **one real bank account**. The system tracks each fund's share of it and never lets the total bank balance go below zero.

How a transaction flows

1. Enter Accountant fills Form 203 → **2. Check** System checks every rule → **3. Submit** Sent to office head → **4. Approve** Gets गो.भौ.नं. → **5. Books** All books update by themselves

Main features at a glance

Entry

Form 203 voucher (easy payment mode or full debit/credit mode), income entry, tax deposit voucher, payee register, budget sub-heads.

Control

Approval workflow, budget check, bank-balance check, period locks, date-order numbering, duplicate-bill warning, role permissions.

Books

बैंक नगदी किताब (per fund and consolidated), हिसाब विवरण, TDS pending list, income by source, dashboards with charts.

Trust

Tamper-proof audit log (hash-chained and blocked from editing at database level), exact money (paisa, no rounding errors), Nepali/English, Nepali dates (BS).

2. Users and what each one can do

There are three roles. Every login belongs to one role, and the server checks the role on every action, so hiding a button is never the only protection.

Action	Super Admin (कार्यालय प्रमुख)	Accountant (लेखापाल)	Auditor (लेखापरीक्षक)
See all funds, vouchers, books, reports, charts	Yes	Yes	Yes
Print vouchers and books	Yes	Yes	Yes (logged)
Create and edit draft vouchers, income entries, payees	Yes	Yes	No
Submit a voucher for approval	Yes	Yes (own vouchers)	No
Cancel (रद्द) a draft or submitted voucher	Yes	Yes (own vouchers)	No
Approve or reject a voucher	**Yes (only this role)**	No	No
Approve spending above budget (with written reason)	Yes	No	No
Mark a voucher as paid (भुक्तानी भएको)	Yes	Yes	No
Settings: organisation, budget, heads, sub-heads, users, tax rates, opening balance, voucher numbering, month locks	Yes	No	No
Consolidated cash book (all four funds)	Yes	No	Yes
Audit log and tamper check	Yes	No	Yes

Demo logins

Username	Password	Role
admin	`Admin12345`	Super Admin
lekha	`Lekha12345`	Accountant
auditor	`Audit12345`	Auditor

> **Login safety.** After **5 wrong passwords** an account is locked for **30 minutes**. A session also ends by itself after **30 minutes without activity** (changeable in Settings). Change all demo passwords before entering real office data.

3. Starting, stopping and sharing the site

The project folder `C:\chetra 2` has three double-click files:

File	What it does
start.bat	Starts the website and its own database (one minimised window called "Pathibhara Accounts – do not close"). It waits until the site answers, then opens `http://localhost:3000` in the browser.
share.bat	Creates a free public link (`https://….trycloudflare.com`) so anyone on the internet can open the site. It prints the link and copies it to the clipboard. The link works only while this PC is on, connected, and the window stays open. A new link is made every time.
stop.bat	Stops the share link, the website and the database.

Ways to reach the site

Who	Address
On the same PC	`http://localhost:3000`
Other PCs in the office (same network)	`http://<this-PC-name>:3000`
Anyone on the internet	The link printed by share.bat
Permanent cloud link (optional)	Render.com using the ready-made `render.yaml` . It works even when the PC is off. The free plan sleeps after 15 minutes idle, and its free database expires after 30 days.

> Keep the office PC from sleeping (Windows Settings → System → Power → Sleep: Never) while it acts as the server.

4. Map of every screen

Navigation goes **Home** → **आन्तरिक/अनुदान** → **पूँजीगत / चालु / आम्दानी**, and then the tabs inside a fund.

Screen	Nepali name	What it shows / does	Who
Home	गृहपृष्ठ	Summary tiles (shared bank balance, awaiting approval, TDS to deposit, days left in FY), spending-by-fund chart, and the आन्तरिक / अनुदान cards	All
Source page	आन्तरिक or अनुदान	Three cards: पूँजीगत खर्च, चालु खर्च, आम्दानी	All
Fund → dashboard	ड्यासबोर्ड	Budget, spent, remaining, status counts, money-flow chart	All
Fund → new voucher	भौचर प्रविष्टि	Form 203 entry (easy payment or debit/credit mode)	Admin, Accountant
Fund → voucher list	भौचर सूची	Search, filter by status, open a voucher to print / submit / approve / reject / cancel / mark paid	All
Fund → cash book	बैंक नगदी किताब	Bank-cash book with running balances and monthly totals	All
Fund → tax	कर (TDS)	Pending tax and the कर जम्मा voucher	All (create: Admin, Accountant)
Fund → budget statement	हिसाब विवरण	Allocated / spent / saving per head	All
Fund → next phase	बजेट खाता · पेस्की · प्रतिवेदन	Placeholders for coming features	All
Income	आम्दानी	Income by head, new income entry, income voucher list	All (entry: Admin, Accountant)
Consolidated	सबै कोष (एकीकृत)	Cash book of all four funds together: the true bank balance	Admin, Auditor
Audit	लेखाजोखा अभिलेख	Every action logged, with a tamper check	Admin, Auditor
Settings	सेटिङ	संस्था · बजेट उपशीर्षक · बजेट · सुरुको मौज्दात · भौचर नम्बर · अवधि बन्द · प्रयोगकर्ता · कर दर · खाता/शीर्षक · भुक्तानी पाउने (details in chapter 5)	Admin
Top bar	—	Fiscal year · नेपाली/English · Devanagari/Latin digits · light/dark theme · logout	All

5. First-time setup (Super Admin)

Do these once, in this order, before the accountant starts entering vouchers. Everything is under **सेटिङ (Settings)**.

#	Tab	What to enter	Why it matters
1	संस्था	Office name, government level, ministry, office code, PAN, address, the names and posts of the लेखा प्रमुख and कार्यालय प्रमुख	Printed on Form 203, the cash book and statements
2	खाता/शीर्षक	Check the expense heads (code, क्रियाकलाप नं., name, पूँजीगत/चालु) and the income heads (code, आन्तरिक/अनुदान). Add the bank account name, branch and number.	48 expense heads and 11 income heads are pre-loaded with **provisional** codes. Correct them before first use; a head's code is frozen once a voucher uses it.
3	बजेट उपशीर्षक	Budget sub-head numbers and names from the red book (रातो किताब)	Chosen on each voucher
4	बजेट	Appropriation (विनियोजन) per head per fund. Later changes are added as: थप बजेट (additional), बजेट कटौती (reduction) or रकमान्तर (transfer between heads)	Each change is a separate row with a note, so the history is kept. The budget of a head is the sum of its rows.
5	सुरुको मौज्दात	Opening bank and cash balance of each fund on 2082.04.01	The first row of the cash book. Cannot be changed once any voucher is approved.
6	भौचर नम्बर	If paper books already reached, say, voucher 159, set the next number to 160	Can be set only once, before the first approval
7	प्रयोगकर्ता	Create real users, give roles, set strong passwords, deactivate demo users	Approval rights depend on role
8	कर दर	Confirm TDS rates (have a Chartered Accountant check them)	A new rate is added with an effective date; old vouchers keep the old rate
9	भुक्तानी पाउने	Register regular suppliers with PAN (9 digits), VAT status and bank account	Needed for TDS, EFT and the tax register

6. Daily work: entering a payment voucher

Go to **Home → आन्तरिक or अनुदान → पूँजीगत or चालु → भौचर प्रविष्टि**. There are two modes.

A. सजिलो भुक्तानी (easy payment mode): used for almost everything

1. **Date (मिति)** in BS, format `YYYY.MM.DD`. It must be inside the open fiscal year and in an unlocked month.
2. **Budget sub-head** and project name (optional).
3. **खर्च शीर्षक (expense head)**: the list shows only heads of this fund's type (capital heads in पूँजीगत funds, recurrent heads in चालु funds) and the **remaining budget** next to each. The खाता पाना नं. and क्रियाकलाप नं. fill in automatically from the head.
4. **Payee (भुक्तानी पाउने)**: pick from the register or add a new one on the spot.
5. **Bill amount, bill number, bill date**. Tick "*बिलमा १३% मू.अ.कर समावेश छ*" if the bill includes VAT. This is allowed only for VAT-registered payees.
6. **Tax type**: the system calculates TDS and the net payment instantly (see chapter 8). To change the TDS, tick "TDS परिवर्तन गर्नुहोस्" and write a reason.
7. Several payees can be added to one voucher.
8. **Payment method**: EFT (needs the payee's bank account) or cheque.
9. **व्यहोरा (narration)**: a short description is required.
10. **Preview** shows the exact Form 203 as it will print. Then **save as draft** or **submit**.

The system builds the accounting lines itself:

```
Dr  Expense head ..... bill amount (gross)
    Cr  Bank ............... net payment (bill − TDS)
    Cr  TDS payable ......... tax withheld
```

B. डेबिट/क्रेडिट (advanced mode)

For adjustments and unusual entries, the accountant writes each Dr/Cr line by hand, with the source columns (स्रोत व्यहोर्ने निकाय, भुक्तानी विधि). Total debit must equal total credit, and the screen shows the difference live.

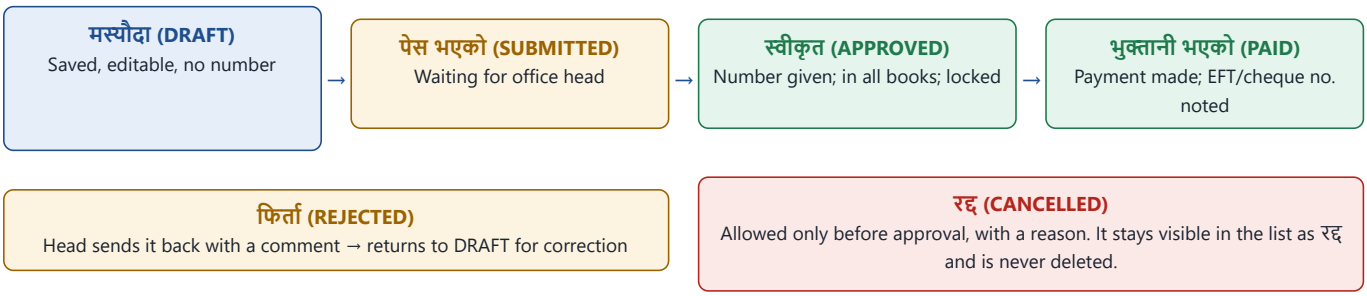
Warnings you may see while entering

- **दोहोरो बिल हुनसक्छ (possible duplicate):** the same payee and bill number already exist, or the same payee and amount on the same day. You must tick "मैले जाँचें, यो दोहोरो होइन" to continue.
- **Budget will be exceeded:** shown at submit. Only the Super Admin can approve it, and only with a written reason, which is recorded.

Printing Form 203

Open the voucher and press **प्रिन्ट**. The page is an exact copy of the paper म.ले.प. फाराम नं. २०३ on one A4 sheet, including the उद्देश्य and फाराम भर्ने तरिका text, the amount in words, and the signature blocks with names and BS dates. Each print is recorded in the audit log.

7. Voucher life: draft → submit → approve → paid



What happens at each step

Step	Who	What the system checks or does
Save draft	Accountant / Admin	All structural rules (dates, heads, PAN, Dr = Cr, VAT rules, duplicate warning). A draft can be edited any number of times.
Submit	The person who prepared it (or Admin)	The month must be open. The **budget check** runs: if any head would go over budget it stops, unless the Super Admin gives a reason.
Approve	**Super Admin only**	1. Only one approval at a time runs for the whole office, so numbers and balances never clash. 2. Date order: a voucher cannot be approved with a date earlier than the last approved voucher of that fund, so numbers stay in date order. 3. The budget check is repeated. 4. Bank check: the shared bank account (all four funds together) can never go negative. 5. For a tax deposit voucher, the amount must equal the TDS rows it clears. 6. Gives the next **गो.भौ.नं.** for that fund and year: sequential, gap-free, never reused. 7. From now on the voucher appears in every book.
Reject	Super Admin	Comment required. The voucher goes back to DRAFT with the comment shown at the top.
Cancel	Preparer / Admin	Only DRAFT or SUBMITTED. Reason required. An approved voucher can never be cancelled or edited; a correction is made with a new समायोजन (adjustment) voucher.
Mark paid	Accountant / Admin	Records that the bank payment was actually made, with the EFT/cheque reference.

> Two people editing the same voucher at once is detected: the second save is refused with "someone changed this voucher, reload" (version check), so no change is silently lost.

8. Tax deducted at source (TDS) and कर जम्मा

How the tax is calculated

1. If the bill includes 13% VAT, the taxable base = bill ÷ 1.13. Otherwise the base = bill.
2. TDS = base × rate of the chosen tax type. The rate used is the one **in force on the voucher date**.
3. Rounding follows the office setting. The default is **up to the next whole rupee**, which matches the client's books.
4. Net payment = bill − TDS.

All of this uses exact fractions, never decimals, so there is never a one-paisa error.

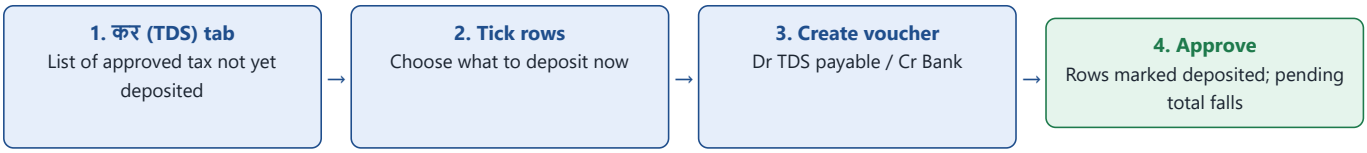
Tax type (default rates: verify with a CA)	Rate	Base
मू.अ.कर दर्ता आपूर्तिकर्ता/सेवा (VAT_SUPPLIER)	1.5%	amount excluding VAT
मू.अ.कर दर्ता नभएको फर्मको बिल (NON_VAT_FIRM)	1.5%	bill
व्यक्तिलाई सेवा/परामर्श शुल्क (INDIVIDUAL_SERVICE)	15%	bill
तलब/ज्याला सामाजिक सुरक्षा कर (SALARY_SST)	1%	bill
घर/सवारी भाडा (RENT)	10%	bill
कर छुट (EXEMPT)	0%	reason required

Worked example: the client's voucher 160

```
VAT bill (परामर्श सेवा) ..............  79,100.00
Base = 79,100 ÷ 1.13 ..................  70,000.00
TDS  = 70,000 × 1.5% ..................   1,050.00
Net payment ...........................  78,050.00

Lines:  Dr परामर्श सेवा 79,100 | Cr Bank 78,050 | Cr TDS payable 1,050
Bank: 5,03,09,038.73 (opening) − 78,050 = 5,02,30,988.73  ✓ matches the paper cash book
```

Depositing the tax (कर जम्मा)



A row being deposited cannot be picked twice. If the deposit voucher is rejected or cancelled, the rows return to the pending list.

9. Income (आम्दानी)

Go to **Home** → **आन्तरिक or अनुदान** → **आम्दानी** (the green card). The page has three parts:

1. **स्रोत अनुसार आम्दानी:** every income head of that source with the amount received (approved), the amount awaiting approval, a share bar and the total.
2. **नयाँ आम्दानी प्रविष्टि:** the entry form.
3. **आम्दानी भौचरहरू:** every income voucher with its number, date, head, payer, receipt number, fund and status. Click the number to open or print it as Form 203.

Income heads loaded (provisional codes)

आन्तरिक (own revenue)

14211 प्रवेश/टिकट शुल्क · 14213 होटल, लज तथा पसल भाडा · 14212 सेवा शुल्क · 14229 पार्किङ शुल्क · 14311 दान, भेटी तथा चढावा · 14111 बैंक ब्याज · 14529 विविध आम्दानी

अनुदान (grants)

13311 संघीय सरकार अनुदान · 13312 प्रदेश सरकार अनुदान · 13313 स्थानीय तह अनुदान · 13314 अन्य संस्था/दातृ निकाय अनुदान

The Super Admin can add more under **सेटिङ** → **खाता/शीर्षक**: choose type **REVENUE** and pick आन्तरिक or अनुदान.

Entering income

1. मिति, and कोष: which fund (चालु or पूँजीगत) the money is for.
2. आम्दानी शीर्षक: only heads of this source are listed.
3. रकम; and where it went: बैंक खाता or नगद (cash).
4. रकम बुझाउने व्यक्ति/निकाय (who paid) is required; रसिद नं. (receipt number) is optional.
5. व्यहोरा: left empty, it becomes "<head> बापत प्राप्त रकम".
6. **पेस गर्नुहोस्** creates and submits it. The office head approves it like any voucher and it gets the next गो.भौ.नं. of that fund.

```
Dr  Bank (or Cash) ..... amount
    Cr  Income head ..... amount
```

- Once approved, the bank balance in the cash book rises. The income is **not** counted as spending in the हिसाब विवरण.
- The fund's chart shows an extra "आम्दानी" line.
- An अनुदान head cannot be used in an आन्तरिक fund, and the other way round. The system blocks it.
- On the printed Form 203 the payer and receipt number are added to the narration, and the payment table stays empty (nobody is paid).

10. Books and reports produced automatically

Every book below is **calculated** from approved vouchers each time it is opened, so it is never stored separately and never goes out of date. Each book has a date range filter (from/to, BS) and prints on A4.

बैंक नगदी किताब (Bank-Cash Book)

Column	Filled from
मिति, गो.भौ.नं., विवरण, चेक नं.	Voucher date, number, narration, EFT/cheque reference
नगद मौज्दात: डेबिट / क्रेडिट / बाँकी	Lines on the cash account, with running balance
बैंक: डेबिट / क्रेडिट / बाँकी	Lines on the bank account, with running balance
बजेट खर्च	Net of expense-head lines
पेस्की: दिएको / फछ्र्यौट	Advance account lines
विविध: डेबिट / क्रेडिट	Everything else (TDS payable, income heads, deposits)

It starts with the **मौज्दात जिम्मेवारी** (opening) row, and for each month shows **मासिक जम्मा** (month total) and **आगामी महिना सारेको मौज्दात** (balance carried forward), then the yearly total and closing balance. The **consolidated** version (सबै कोष) merges all four funds and shows the true balance of the one shared bank account.

हिसाब विवरण (Budget Statement)

For each head: **विनियोजित** (sum of all budget rows) − **जम्मा खर्च** (approved spending) = **बचत** (saving). बचत is always calculated, never typed.

Dashboards and charts

- **Home:** shared bank balance, vouchers awaiting approval, TDS to deposit, days left in the year. Each source card shows its budget, spending and %. The chart shows spending per fund.
- **Fund dashboard:** budget, spent, remaining, counts by status, rejected vouchers. The money-flow chart has lines for budget expense, bank payment, TDS and income. You can view it by month for the whole year, by week for one month, or as cumulative spending against the budget line. Every chart also has a table view.
- **Source page:** capital, recurrent and income cards with totals.

Acceptance check against the client's paper books

The client's figures were entered and every running balance matched to the paisa (opening 5,03,09,038.73; V160 → 5,02,30,988.73; V161 → 5,02,29,938.73; V162 → 4,97,83,480.73 …). These are kept as automatic tests (chapter 17).

11. Monthly and year-end routine

Every day

- Accountant enters and submits vouchers and income. The head approves or rejects them from the भौचर सूची (filter "पेस भएको"). After paying, mark vouchers भुक्तानी भएको.

Every month

1. Check that no voucher of the month is still DRAFT or SUBMITTED.
2. Deposit the month's TDS from the कर (TDS) tab.
3. Print the month's cash book page and compare the bank balance with the bank statement.
4. **Lock the month** (सेटिङ → अवधि बन्द). After locking, nobody can create, submit or approve a voucher dated in that month. Only the Super Admin can unlock it, and both actions are logged.

Year end (Ashadh end)

1. Clear all drafts and pending vouchers. The year cannot be closed while any remain.
2. Print the yearly cash book, हिसाब विवरण and audit log.
3. Create the next fiscal year and enter its opening balance (the closing balance of this year).
4. Close the old year.

12. Every safety rule and error message

All rules run on the server, whatever screen or tool is used. The messages appear in Nepali or English.

Rule	When it stops you
Debit = Credit	Totals differ on any voucher
Amounts positive	Zero or negative line amount (also enforced inside the database)
Budget	Spending would exceed the head's remaining budget; only the Super Admin may override, with a reason of at least 5 characters
Bank never negative	Approval would take the shared bank (or cash) below zero
Head type	Capital head in a चालु fund or recurrent head in a पूँजीगत fund
Income source	अनुदान income head in an आन्तरिक fund or the other way round
Date	Invalid BS date, a date outside an open fiscal year, or a date in a locked month
Date order	Approving a voucher dated before the fund's last approved voucher
PAN	Not exactly 9 digits
EFT	Payee has no bank account
VAT	VAT ticked for a payee not registered for VAT
TDS change / exemption	No reason given; changed TDS below 0 or above the bill
Duplicate	Same payee and bill no., or same payee and amount on the same day (needs confirmation)
Narration	Empty व्यहोरा
Approved = final	Editing or cancelling an approved voucher
Head in use	Changing the code or type of a head or sub-head already used in a voucher (the name can still be corrected)
Opening balance	Changing it after any approval; an opening balance on an expense or income head
Database down	"डाटाबेस चलिरहेको छैन": restart with start.bat (the server also restarts its database by itself every 10 seconds)

13. Audit trail and security

- **Everything is logged:** logins (including failed ones), creating, editing, submitting, approving, rejecting, cancelling and printing vouchers, and every settings change. Each entry records who, when, from which computer, and the data before and after.
- **Nobody can change the log.** A database trigger blocks UPDATE, DELETE and TRUNCATE on the log table, even for the database administrator or a bug in the program.
- **Tamper check:** each entry stores a fingerprint (SHA-256 hash) of itself plus the previous entry, forming a chain. **लेखाजोखा अभिलेख → जाँच गर्नुहोस्** recomputes the chain and shows either "सुरक्षित छ" or the exact entry that was altered.
- **Passwords** are stored only as bcrypt hashes. Login tokens are stored as SHA-256 hashes, so a stolen database cannot be used to log in.
- **Auditor** is read-only everywhere; the server refuses any change from that role.
- **Money** is stored as whole paisa (BIGINT). There are no floating-point numbers anywhere, so totals are always exact.

14. Technical design (for developers)

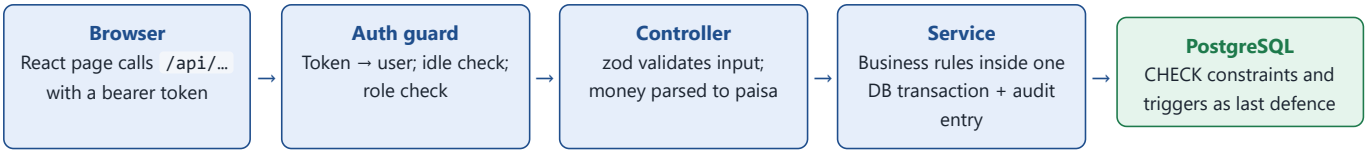
Technology

Layer	Technology
Browser app	React 18 + TypeScript, Vite, Tailwind CSS, TanStack Query, react-router, react-i18next (ne/en), offline fonts
Server	NestJS 11 (Node.js 22+), zod validation, bcryptjs; it also serves the built browser app
Database	PostgreSQL (UTF-8) through Prisma 6. In single-PC mode (`EMBEDDED_PG=1`) the server starts its own PostgreSQL on port 5544.
Shared core	`@pa/core` : money, TDS, BS calendar, amount in words, voucher rules. The same code runs in the browser and on the server, so the preview always equals what is saved.

Folder layout

```
C:\chetra 2
├── start.bat · stop.bat · share.bat     double-click helpers (chapter 3)
├── render.yaml                          one-click cloud deployment
├── packages/core/src                    @pa/core — shared accounting core
│   ├── money.ts      parse/format paisa, Nepali digit grouping (5,03,09,038.73), Devanagari digits
│   ├── tax.ts        computeTds(): exact-fraction TDS with rounding modes
│   ├── bsdate.ts     BS ⇄ AD conversion, today in Nepal time, fiscal-year helpers
│   ├── words.ts      amount in words (नेपाली / English)
│   └── voucher.ts    line validation, payment/tax-deposit line builders, cash-book columns, PAN check
├── apps/server
│   ├── prisma/schema.prisma            database model (chapter 15)
│   ├── prisma/migrations/              init · integrity guards · head activity & paid · income
│   ├── prisma/seed.ts, seed-demo.ts    empty office / demo data
│   ├── scripts/add-heads.ts            add new standard heads to an existing office
│   ├── test/acceptance.e2e.test.ts     31 end-to-end tests with client figures
│   └── src
│       ├── main.ts, bootstrap.ts       start-up, .env, embedded DB watchdog, serve web app
│       ├── auth/                       login, sessions, idle timeout, lockout, role guard
│       ├── audit/audit.service.ts      hash-chained audit log
│       ├── master/                     settings: org, years, locks, heads, sub-heads, payees, tax, budget, opening, users, audit
│       ├── vouchers/vouchers.service.ts  compose + validate, submit, approve, reject, cancel, paid, tax deposit
│       ├── vouchers/ledger.ts          balance/budget queries from approved lines
│       ├── reports/reports.service.ts  cash book, budget statement, trend, income, fund summaries
│       ├── common/                     errors, DB-down handling, zod helpers, BigInt→JSON
│       └── seed/seed.ts                48 expense heads, 11 income heads, tax rules, 4 funds
└── apps/web/src
    ├── App.tsx                         routes (chapter 4)
    ├── pages/  Home · VoucherEntry · Vouchers · Income · Reports · Admin
    ├── components/  Form203 (exact paper form) · BsDateInput · PayeeForm · TrendChart · Layout · ui
    ├── i18n/ne.ts, en.ts               every label and error message
    └── lib/  api · session · prefs · types
```

How a request travels



Key design decisions

- **Books are derived, not stored.** `ledger.ts` sums approved voucher lines on demand. Nothing can drift out of sync.
- **One compose() function** builds the lines for preview, create and update, so what is previewed is exactly what is saved.
- **Money is BIGINT paisa**, sent over JSON as exact decimal strings ("78050.00").
- **Approvals are serialised** with a PostgreSQL advisory lock per office, and the voucher counter row is locked (`FOR UPDATE`), which gives gap-free numbers even with many users.
- **Optimistic locking:** each voucher has a version number, and stale edits are refused.
- **BS dates** are stored both as text (`2082.04.01`) and as an AD date for sorting and ranges.

15. Database tables

Table	Holds
Organization	Office details, signatories, VAT rate, TDS rounding, session timeout
User, Session	Logins, roles, lockout counters; active sessions (hashed tokens)
FiscalYear, PeriodLock	Fiscal years (BS and AD start/end, closed flag); locked months
Fund	The four funds (source × nature)
LedgerAccount	Chart of accounts. Kind = EXPENSE (with nature), REVENUE (with source), BANK, CASH, TDS_PAYABLE, ADVANCE, DEPOSIT, OTHER. Also holds code, क्रियाकलाप नं., खाता पाना नं. and bank details.
BudgetSubHead	बजेट उपशीर्षक numbers
BudgetEntry	One row per budget change (appropriation, addition, reduction, transfer pair)
OpeningBalance	मौज्दात जिम्मेवारी per fund and account
Payee	Payee register (PAN, VAT, bank)
TaxRule	TDS rate history with effective dates
VoucherCounter	Next गो.भौ.नं. per fund and year
Voucher	Header: fund, year, number, type, mode (PAYMENT / MANUAL / TAX_DEPOSIT / RECEIPT), status, dates, narration, total, who prepared/submitted/approved/cancelled/paid and when, version
VoucherLine	Dr/Cr lines: account, kind, amount, activity, component, source columns
VoucherPayee	Payment table rows: payee, PAN, bank, bill, gross, TDS, net, link to the tax deposit voucher (for income: payer and receipt no.)
Attachment	Files attached to vouchers (bills, quotations)
AuditLog	Append-only, hash-chained history of every action

Database-level guards: amounts > 0; gross − TDS = net; an approved voucher must have a number; tax rate 0–100%; month 1–12; audit log immutable.

16. API reference

All paths start with `/api` and need `Authorization: Bearer <token>`, except login. Write actions are refused for the Auditor role.

Method and path	Purpose	Role
`POST /auth/login · POST /auth/logout · GET /auth/me`	Sign in / out, current user	all
`PATCH /auth/me/language · /auth/me/password`	Own language and password	all
`GET /meta`	Office, funds, years, today (BS), sub-heads, tax rules	all
`PATCH /organization`	Office settings	Admin
`POST /fiscal-years · POST /fiscal-years/:id/close`	New year / close year	Admin
`GET, POST /fiscal-years/:id/locks`	Month locks	Admin (GET all)
`GET /accounts · POST /accounts · PATCH /accounts/:id`	Chart of accounts (expense and income heads, bank…)	Admin (GET all)
`POST /sub-heads · PATCH /sub-heads/:id`	Budget sub-heads	Admin
`GET, POST /payees · PATCH /payees/:id`	Payee register	Admin, Accountant
`GET, POST /tax-rules`	TDS rates	Admin (GET all)
`GET, POST /budget-entries`	Budget rows	Admin (GET all)
`GET, POST /opening-balances`	Opening balance	Admin (GET all)
`GET, POST /users · PATCH /users/:id`	Users	Admin
`GET /audit · GET /audit/verify · POST /audit/print`	Audit log, tamper check, log a book print	Admin, Auditor
`GET /funds/:fundId/vouchers`	Voucher list (status, search, pages)	all
`POST /funds/:fundId/vouchers/preview`	Exact Form 203 preview	Admin, Accountant
`POST /funds/:fundId/vouchers`	Create draft (mode PAYMENT, MANUAL or RECEIPT)	Admin, Accountant
`GET /funds/:fundId/budget-position`	Budget, spent, pending, remaining per head	all
`GET /funds/:fundId/tds/pending · POST /funds/:fundId/tax-deposit`	Pending TDS; create tax deposit voucher	all / Admin, Accountant
`POST /funds/:fundId/voucher-start-number`	Starting गो.भौ.नं. (once)	Admin
`GET /vouchers/:id · PUT /vouchers/:id`	Read / edit draft	all / Admin, Accountant
`POST /vouchers/:id/submit · /cancel · /paid`	Workflow	Admin, Accountant
`POST /vouchers/:id/approve · /reject`	Workflow	Admin
`POST /vouchers/:id/printed`	Record a print	all
`GET /reports/cash-book/:fundId · GET /reports/cash-book`	Cash book per fund / consolidated	all / Admin, Auditor
`GET /reports/budget-statement/:fundId`	हिसाब विवरण	all
`GET /reports/income/:source`	Income by head + income vouchers	all
`GET /reports/trend · GET /reports/fund-summaries`	Chart data; dashboard cards	all

17. Testing

Run `npm test` from the project folder. Each run creates a fresh database, so tests never touch real data.

Suite	Count	Covers
Core unit tests	21	Money parsing and formatting, TDS rounding, BS calendar, amount in words, voucher rules
Client figures (§12)	12	Opening balance, V160–V183 TDS and net, running bank balances, budget savings, charts equal books, cash book reconciles to the paisa
No-mistake rules (§8)	10	Dr ≠ Cr, overspend + override, bank overdraft, head type, dates, locked month, duplicates, VAT/TDS reasons, PAN, date order
Roles and audit (§3)	7	Auditor read-only, accountant limits, approved is final, lockout, immutable log and intact chain, sub-heads, reject flow
Income	2	Dr bank / Cr income, source mismatch blocked, pending/received totals, bank rises, not counted as expense; only Admin adds income heads

> Last run: **all 31 server tests and 21 core tests passed.**

18. Troubleshooting

Problem	Fix
"This site can't be reached" at localhost:3000	Double-click start.bat and wait for the browser to open.
"डाटाबेस चलिरहेको छैन" on login	Run stop.bat, then start.bat.
Public link does not open	The PC slept or share.bat was closed. Run start.bat, then share.bat, and send the **new** link.
"Account locked"	5 wrong passwords. Wait 30 minutes, or the Super Admin unlocks it in सेटिङ → प्रयोगकर्ता.
Logged out by itself	30 minutes without activity (session timeout). Log in again.
Cannot approve: date order	A later-dated voucher of this fund is already approved. Correct the date, or approve vouchers in date order.
Cannot approve: budget	Add budget (थप बजेट / रकमान्तर), or approve as Super Admin with a written reason.
Cannot approve: insufficient balance	The shared bank does not have enough money. Record the income or grant first.
Cannot enter a date	The month is locked, or the date is outside the fiscal year.
A head is missing from the list	It belongs to the other type (capital/recurrent, internal/grant) or is inactive. Check सेटिङ → खाता/शीर्षक.

19. What is still to be built, and what must be verified

Done

Schema, roles, login, audit trail · four-fund navigation · Form 203 entry with TDS engine and approval workflow · bank-cash book, budget statement and printing · tax deposit · dashboards and charts · bilingual interface · income (आम्दानी) · all acceptance tests.

Next phases

Feature	What it adds
Backup / restore	One-click backup of all data to a file and restore. **Recommended before go-live.**
पेस्की (Advance)	Giving and settling advances; advance register with ageing
बजेट खाता (Budget ledger)	Page for each head with every voucher and its running balance
TDS register and IRD monthly report	Tax withheld per payee and PAN, ready for filing
Excel export	Cash book, हिसाब विवरण and income report as .xlsx
Bank reconciliation, trial balance, other reports	Month-end checks
Licensing	Licence key per installation

Must be verified before go-live

1. **Expense-head codes and क्रियाकलाप नं.** (48 heads) against the office's approved budget book.
2. **Income-head revenue codes** (11 heads) against the office books.
3. **All TDS rates**, by a Chartered Accountant against the current Finance Act.
4. **Client book findings:** the TDS on paper vouchers V165 (8,884) and V185 (456) matches no standard rate. The हिसाब विवरण shows less spending than the cash book for परामर्श, पर्व मेला, सेवा करार, बैठक भत्ता and फर्निचर. Capital heads appear in the चालु statement. The paper books shared are FY 2081/82.
5. **Replace all demo passwords** and deactivate the demo users.

End of guide.